

Exp No: 13: EMBEDDED C PROGRAM TO ADD AN ARRAY OF 16-BIT NUMBERS AND STORE THE 32-BIT RESULT IN INTERNAL RAM

PROGRAM:

```
#include <reg51.h>

void main(void)
{
    unsigned int i;

    array[00x00 unsigned int array[5] = {0x1111, 0x2222, 0x8888, 0x4444, 0xABCD};

    unsigned long sum = 0;

    for (i = 0; i < 5; i++)
    {
        sum = sum + array[i];

        P0 = (unsigned char)(sum & 0xFF);

        // Lower 8 bits

        P1 = (unsigned char)((sum >> 8) & 0xFF); // Next 8 bits

        P2 = (unsigned char)((sum >> 16) & 0xFF); // Upper 8 bits

        while(1); // keep output stable
    }
}
```

Output :

